CAS HOMEOWNERS PRICING CASE STUDY DETAILS

Case Overview

You are the team lead for Alpha Actuarial (Alpha), a consulting firm specializing in homeowners' insurance. Your current client is a mid-size insurance firm, Pickles Mutual Insurance Company (Pickles). Pickles has underwritten homeowner's insurance and produced artisan pickles since 1835. Pickles is in a pickle and is looking for assistance in understanding their homeowner's book of business.

Pickles has collected data on some information which may be useful in pricing homeowner's insurance for their portfolio of ~200,000 insureds. Pickles has engaged Alpha to perform an actuarial analysis and make recommendations on the discount structure.

Relevant Industry Information

Rating by Peril

Homeowners' insurance is rated by peril at Pickles. Covered perils include wind & hail, fire, water, theft, other property damage, and liability.

Competition

In recent years, the homeowners' insurance market has become much more competitive. Pickles has a 20% market share. One of the members of your team at Alpha Actuarial has collected information about competitor market share and rating, and it is included in the data.

Credibility

Alpha has done detailed studies on credibility and determined that information is fully credible when there are 1,082 claims.

Data

You have been provided with an Excel workbook that contains premiums, losses and loss adjustment expenses (LAE), and claim counts for the past two years, segmented by types of water sensors, security alarms, and roofing for Pickles.

You have also received competitor data related to your top competitors in your market, as well as projected yearly underwriting expenses.

Deliverables

Your team's goal is to create a pricing structure based on the data collected by Alpha. At the end, you will be presenting your results to the company executives (i.e. "the judges"). The following are some considerations you probably ought to cover in your presentation:

1. Methods: Provide a brief overview of any assumptions you needed to make and mathematical methods used to derive your rating factors. Any methods of determining additional rating factors, including personal experience, are sufficient. Document all your assumptions and why you relied upon your approaches.

2. Results & Insights: Propose a new discount structure. State and explain any assumptions you made or additional considerations you took into account, including the use of competitor

© 2026 by the Casualty Actuarial Society. All rights reserved.

information and the credibility of the data. Include any comments related to how implementing these discounts will impact your existing customers and potential new customers.

3. Additional Considerations: What do you think would be the impact to loss ratio & market share if Pickles made no changes to their rating formula, given that the competitors have discounts? What adjustments would you propose if the CEO wants to target an increase in total premium? What other business considerations should you incorporate as part of your proposal?

4. Communication Style: Your communication should be clear and easy to follow. Be sure to document your assumptions and show your calculations clearly.

Suggested Resources

Basic Ratemaking -
https://www.casact.org/sites/default/files/old/studynotes_werner_modlin_ratemaking.pdf

Glossary of Insurance Terms

Allocated Loss Adjustment Expenses (ALAE) – Claim-related expenses that are directly attributable to a specific claim; for example, fees associated with outside legal counsel hired to defend a claim can be directly assigned to a specific claim.

Combined Ratio – The total of all losses and operating expenses represented as a% of earned premium.

Experience Period – A period of history used by insurance companies to evaluate terms when issuing a policy. Any claim and/or premium amounts within this period will affect the current insurance premium rates the customer receives.

Exposure – An exposure is the basic unit of risk that underlies the insurance premium. The exposure measure used for ratemaking purposes varies considerably by line of business. Some examples are number of vehicles, insured value of building, payroll, square footage, etc. There are four different ways that insurers measure exposures: written, earned, unearned, and in-force exposures.

Incremental Cost – The increase or decrease in costs as a result of one more or one less unit of output.

Indemnity – Compensation for (or to provide against) injury, loss, incurred penalties, or from a contingent liability.

Inflation – A sustained increase in the aggregate or general price level in an economy. Inflation means there is an increase in the cost of living.

Loss Adjustment Expense (LAE) – In addition to the money paid to the claimant for compensation, the insurer generally incurs expenses in the process of settling claims; these expenses are called loss adjustment expenses (LAE). Loss adjustment expenses can be separated into allocated loss adjustment expenses (ALAE) and unallocated loss adjustment expenses (ULAE): LAE = ALAE + ULAE

Loss Costs – Also called "pure premium," the actual or expected cost to an insurer of indemnity payments and allocated loss adjustment expenses (ALAEs). Loss costs do not include overhead costs or profit loadings.

Loss Ratio – The percent of earned premium that goes towards paying claims.

On-level – In this case study, on-level refers to inflation adjustments. Claims costs in the historical

© 2026 by the Casualty Actuarial Society. All rights reserved.

periods are adjusted to today's dollars.

Premium – Is the amount the insured pays for insurance coverage. The term can also be used to describe the aggregate amount a group of insureds pays over a period of time. Like exposures, there are written, earned, unearned, and in-force premium definitions.

Pure Premium – Refer to "loss costs"

Return on Investment (ROI) – Measures the gain or loss generated on an investment relative to the amount of money invested. ROI is usually expressed as a percentage and is typically used for personal financial decisions, to compare a company's profitability or to compare the efficiency of different investments.

Ultimate Costs – Estimate of total costs expected once all claims are settled and paid.

Unallocated Loss Adjustment Expenses (ULAE) – Claim-related expenses that cannot be directly assigned to a specific claim. For example, salaries of claims department personnel are not readily assignable to a specific claim and are categorized as ULAE.

© 2026 by the Casualty Actuarial Society. All rights reserved.